

From Data to Decision: Interpreting Neural Network Outputs

Predicting Student Academic Outcomes Using Artificial Neural Networks

Course: MDSD407 – Artificial Neural Networks (ANN)
Programme: M.Sc. Data Science and Artificial Intelligence

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A. Data Understanding & Preparation

The Problem and the Prediction Task

Student outcome prediction is only useful if it supports intervention early enough to matter. This report predicts which of three outcomes a student reaches, **Dropout**, **Enrolled**, or **Graduate**, using academic, demographic, and socio-economic information. This is a three-class problem, not a simple binary question of whether a student leaves or stays. **Enrolled** is a genuinely unresolved status, and how well the model separates it from the two final outcomes is one of the more informative parts of the analysis.

Dataset Overview

The dataset, **ann-dataset.csv**, describes **4424 students** across **36 input features** plus the target label. Each feature is numeric or integer-coded according to the data dictionary. A check for missing values and duplicate rows returned none, so the dataset appears clean and suitable for modelling without additional data cleaning.

Feature Categories and the Enrollment-Time Question

The 36 features fall into three groups that need to be treated differently by the model. There are **8 binary flags** such as **Debtor** and **Scholarship holder**, which can be used directly. There are **9 nominal categorical columns** such as **Course** and **Nacionality**, whose codes represent categories rather than magnitudes and therefore require one-hot encoding. There are also **19 numeric columns**, which split further into **7 variables known at enrollment** and **12 variables observed only after one or two semesters**, such as counts and grades of curricular units enrolled, approved and evaluated.

This distinction matters more than it first appears. The assignment frames the task as predicting from information known at enrollment, but the semester-performance columns describe how a student is already doing, not who they are at admission. Rather than silently resolve that ambiguity, this report compares two models using the same architecture and procedure.

- **Model A** is restricted to the 24 columns available at enrollment, making it a true early-intervention case.
- **Model B** adds the 12 semester-performance columns, representing a later-stage, higher-accuracy prediction once first-year performance is visible.

Comparing them directly turns the ambiguity in the task framing into an evidenced finding rather than a hidden assumption.

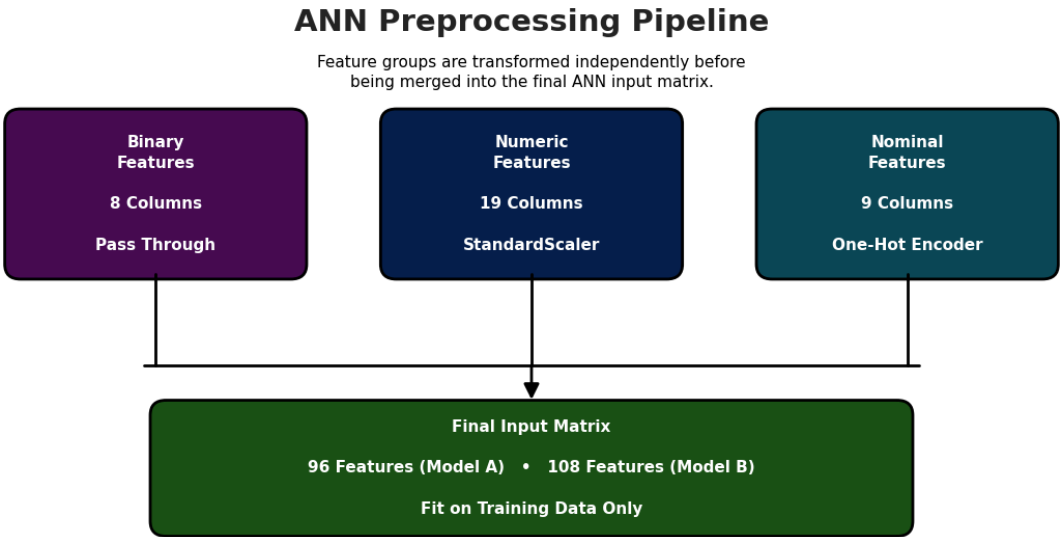


Figure 1. ANN Preprocessing Pipeline

Preprocessing

The dataset was split into **70% training**, **15% validation** and **15% test** sets using stratified sampling to preserve class balance. All preprocessing was fit only on the training split before being applied to validation and test data, which avoids leakage. Binary columns were passed through unchanged, numeric columns were standardized to zero mean and unit variance, and nominal categorical columns were one-hot encoded. Categories appearing in under 1% of the training data were grouped into an **infrequent** bucket, which prevents rare categories from inflating the encoded feature space.

The preprocessing pipeline combined **StandardScaler** for numerical variables, **One-Hot Encoding** for nominal categorical variables, and direct pass-through for binary variables. Separate preprocessing pipelines were maintained for Model A and Model B so that each model received only the features available under its respective prediction scenario.

B. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed using **only the training dataset**, ensuring that no modelling decision was influenced by validation or test data. Rather than examining every variable individually, the analysis focused on patterns with clear relationships to student outcomes and practical relevance for institutional decision-making.

Three key findings emerged:

- **Financial standing** was the strongest enrollment-time indicator of student outcomes. Students with **up-to-date tuition payments, no outstanding debt**, and **scholarship support** were considerably more likely to graduate, while financially vulnerable students showed substantially higher dropout rates. Unlike demographic characteristics, these factors represent areas where institutions can intervene through financial support and student assistance programmes.
- **Age at enrollment** showed a moderate association with academic outcomes. Older students generally exhibited higher dropout rates, suggesting that non-traditional entrants may face additional challenges beyond academic preparedness.
- **Gender** also showed differences across student outcomes. However, this finding is reported as **descriptive rather than actionable**. While gender may be associated with academic outcomes, it should not be used to guide individual intervention decisions.

Overall, the EDA established two important insights that shaped the modelling stage:

- **Financial variables provide the strongest predictive signal available at enrollment and form the basis of the practical recommendations presented later in the report.**
- **Student outcomes are influenced by multiple interacting academic, financial, and demographic factors**, supporting the use of Artificial Neural Networks to learn relationships that may not be captured by simpler linear models.

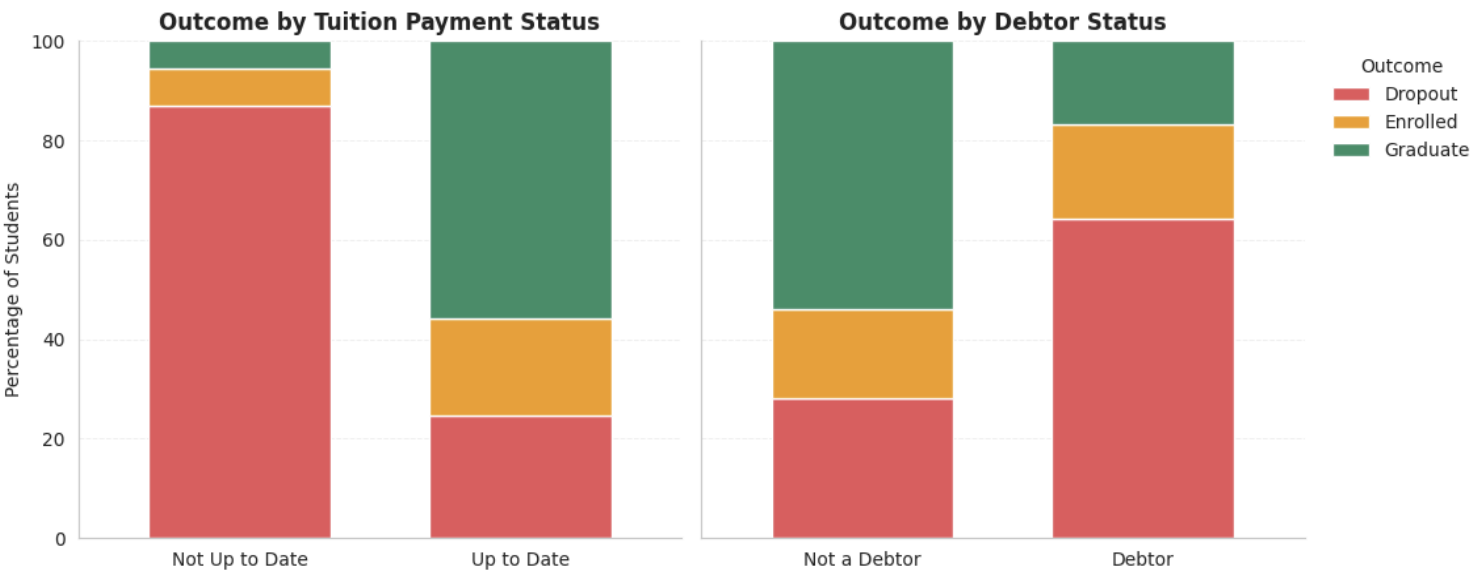


Figure 2. Relationship between Financial Standing and Student Academic Outcomes

C. Neural Network Modeling

A simple feedforward Artificial Neural Network (ANN) was trained separately for each feature set defined in **Section A**, while keeping the architecture and training procedure identical throughout. This ensures that any differences in predictive performance can be attributed to the available input features rather than changes in the modelling approach.

The network architecture was intentionally kept compact, comprising **two fully connected hidden layers** with **ReLU activation**, as the dataset size and problem complexity did not justify a deeper architecture. **Dropout**, **class weighting**, and **early stopping** were incorporated to improve generalisation and reduce the effect of class imbalance during training.

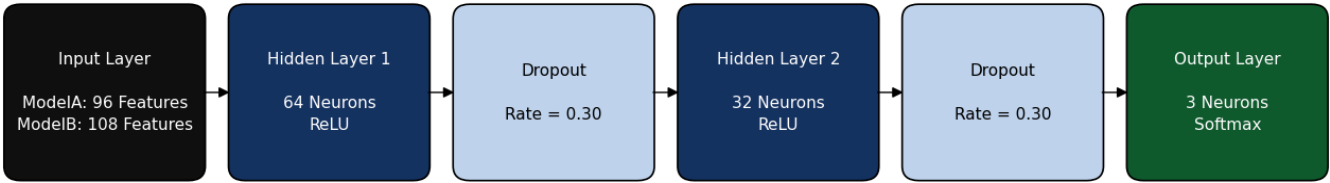


Figure 3. Artificial Neural Network Architecture

The network was formulated as a **three-class classification problem**, using a **Softmax** output layer with **Sparse Categorical Cross-Entropy** loss, which is appropriate for mutually exclusive class labels represented by integer-encoded targets. Model performance was evaluated using **Accuracy**, **Macro Precision**, **Macro Recall**, **Macro F1-score** and **Confusion Matrices**. Since the target classes are moderately imbalanced, **Macro F1-score** was considered alongside Accuracy to provide a more balanced assessment across all three outcomes.

To provide context, two baseline models were also evaluated: a **Majority Class Baseline**, representing the minimum expected performance, and **Logistic Regression**, providing a strong linear benchmark. Model robustness was further assessed across multiple random seeds and train-test splits, while **Permutation Feature Importance** was used to interpret the contribution of individual features.

D. Evaluation & Interpretation

Training Behaviour

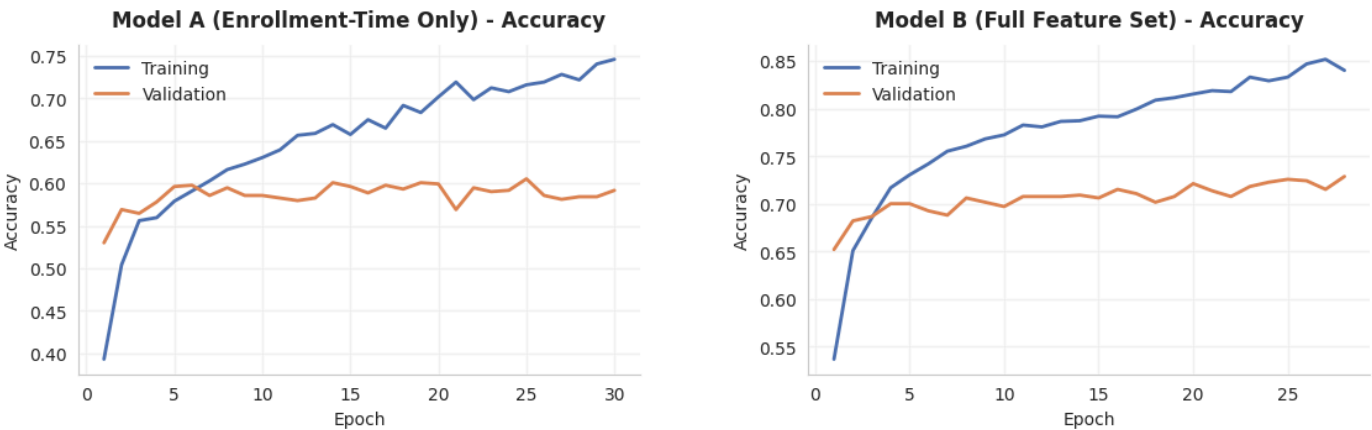


Figure 4. Training and validation accuracy for Model A and Model B

Both models converged steadily during training, with validation performance stabilising within the early-stopping window. Although training loss continued to decrease, validation loss flattened after the initial epochs, showing the start of overfitting and justifying early stopping. **Model B consistently achieved higher validation accuracy than Model A**, suggesting that the additional semester-performance features improved generalisation to unseen data.

Overall Model Performance

Both neural network models substantially outperformed the Majority Class baseline, confirming that they learned meaningful patterns rather than relying on the dominant class. The comparison with Logistic Regression provides the more useful benchmark.

The key observations are:

- **Feature availability had a larger impact than model architecture.** Expanding the feature set from enrollment-time variables (Model A) to the full feature set (Model B) improved both Logistic Regression and the neural network.
- **Model A** remains a practical early-warning model, but its performance is naturally limited by the information available at enrollment.
- **Model B** achieved the strongest overall performance, showing that semester-performance variables add substantial predictive value beyond enrollment-time characteristics.

Model	Raw Features	Encoded Features	Test Accuracy	Macro F1	Enrolled F1
Majority Class	–	–	0.500	0.222	0.000
Logistic Regression (A)	24	96	0.596	0.573	0.415
Model A (ANN)	24	96	0.566	0.534	0.352
Logistic Regression (B)	36	108	0.724	0.689	0.502
Model B (ANN)	36	108	0.748	0.721	0.557

Table 1. Comparison of baseline models and Artificial Neural Networks on the held-out test set

Error Analysis

The confusion matrices show that Model B improves prediction across all three outcome classes compared with Model A, producing more correct classifications and fewer misclassifications overall.

The **Enrolled** class remains the hardest outcome to predict in both models. Students in this group continue to be confused with both **Dropout** and **Graduate**, indicating that they share characteristics with both classes. In contrast, **Dropout** and **Graduate** are more clearly separated, especially once semester-performance variables are introduced.

This is not just a modelling weakness. **Enrolled** is an intermediate status, not a final outcome, so the ambiguity is built into the label itself. The model is therefore capturing a genuine boundary problem rather than a simple feature deficiency.

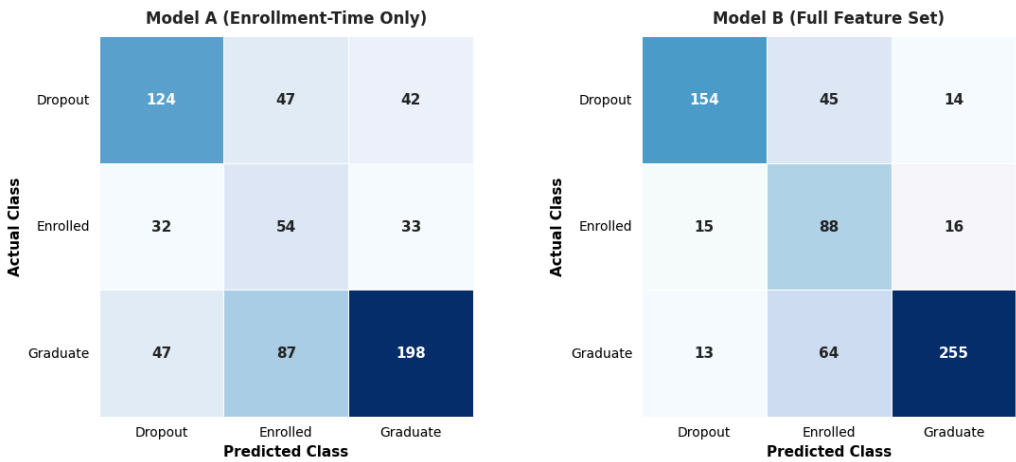


Figure 5. Confusion Matrices for Model A and Model B

Model Robustness and Explainability

The robustness checks show that the overall conclusion remains stable across 5 Seeds and 5 different train-test splits. Individual scores vary slightly, but the relative behaviour stays the same. On the **enrollment-only feature set, neither model has a clear and reliable edge**. On the **full feature set, the neural network’s advantage holds up across both checks**, so the gain is real rather than a one-run artifact. Permutation feature importance explains why.

For Model A, enrollment-time financial variables, especially **Tuition fees up to date**, contribute the strongest predictive signal. Once semester-performance variables are available, academic achievement measures such as **Curricular units approved** become the dominant predictors, while financial variables remain useful but secondary. Together, these results show that **Model B’s improvement comes from the additional academic information**, not from random variation or extra model complexity alone.

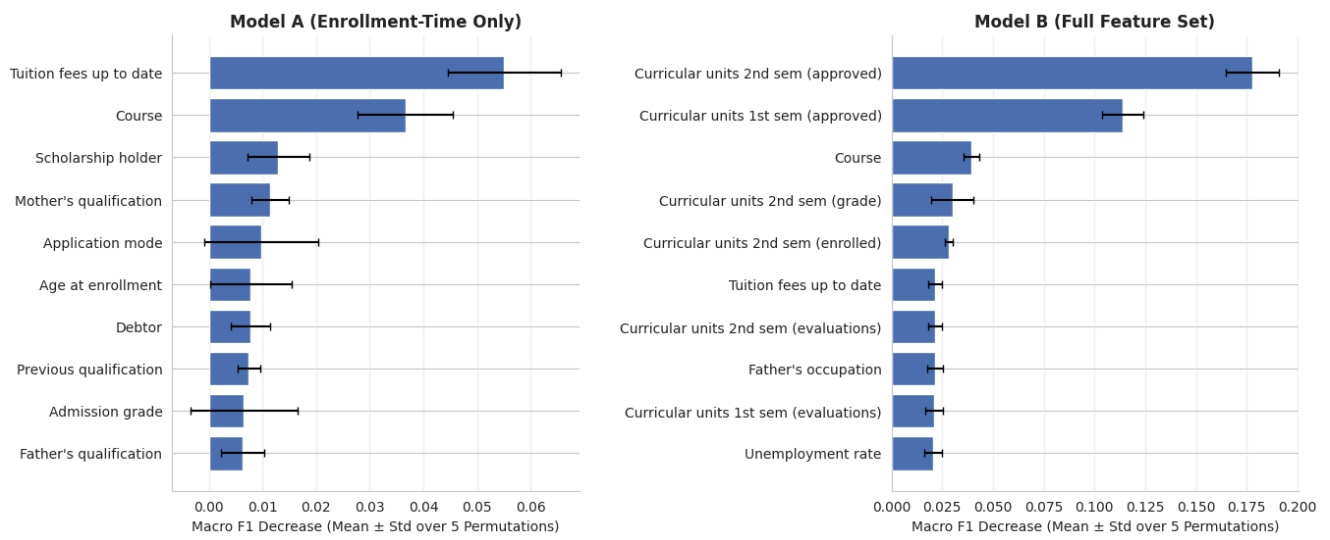


Figure 6. Permutation Feature Importance

E. From Insight to Decision

1. Two-Stage Early-Warning Advising

Use **Model A** at enrollment to flag students into a broad risk tier using only day-one information, and keep the first response lightweight and non-punitive. Once first-semester results are available, re-score students with **Model B** and reserve heavier interventions such as tutoring, mentoring or financial counselling for students flagged by both models.

This two-stage design matters because **Model B is substantially more accurate**, but it only becomes available after a semester has already passed, while **Model A** is weaker but is the only model that can act at the point where intervention is still early.

- **Assumption:** the first contact can be delivered as support, not as a penalty.
- **Risk:** Model A will still produce both false positives and false negatives, especially around the Enrolled/Graduate boundary.
- **Limitation:** age and gender correlate with outcome, but they are not levers. They should inform caution, not individual targeting.
- **Condition for failure:** the model was built on one higher-education system, so thresholds should be re-validated before deployment elsewhere.

2. Targeted Financial-Standing Outreach

Prioritize financial-support outreach for students showing early signs of financial vulnerability, rather than spreading resources evenly across all students. Tuition payment status, debtor status, and scholarship holder status were the strongest actionable predictors in the exploratory analysis, and unlike age or gender, they are factors an institution can actually act on.

- **Assumption:** the observed gaps reflect more than selection effects, so a new intervention should be piloted and measured rather than assumed to work at the same scale.
- **Risk:** tuition and debtor status may also be symptoms of disengagement, not only causes of it.
- **Limitation:** financial support alone will not resolve the Enrolled/Graduate ambiguity seen in the model results.
- **Condition for failure:** the institution must have enough budget and staffing to act on the flags.

Conclusion

This project compared two feedforward Artificial Neural Network models with identical architecture and training, differing only in whether semester-performance information was included. The comparison shows that **enrollment-time information is sufficient for meaningful early-warning prediction**, while **semester-performance variables significantly improve predictive performance without fully resolving the ambiguity of the Enrolled class**. Together, these findings support a practical, staged intervention strategy that combines early identification with more informed decision-making as additional academic information becomes available.